Randall Meyer

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Resume:

Education: Bachelor of Science, Marine Biology, University of Rhode Island, 2002.

Goals: To earn a Ph.D. in the field of Evolution Biology, with special expertise in the study of cultural evolution, meme theory (Humean/Lockean "ideas"), natural language (Neo-Chomskyian Combinatorial Linguistics, Linguistic Typology, LLMs, and Small Language Models, i.e. SLMs), Natural Intelligence (N.I.), Artificial Intelligence (A.I.), robotics, sensory transducers, and analog and digital electronics. More specifically, and to the point, to write the book that I envisioned in 2000-2002, expressing my novel theories regarding these subjects (table of contents and first chapter available upon request). To further the science of robotics and the empirical understanding of humanity, in toto.

Original Hypotheses and Theory:

I. Meyer's Law (Meyer's Combinatorial Linguistic Law), V*(N^2) = S : An expression relating the semantic contents of vocabularies, either Natural or Artificial, to the number of understandable, constructable, simple semantic sentences that can be communicated, or understood, by the intelligences under analysis. One should note, in particular, Chomskyian Linguistics (1957), and perhaps, too, consult, David Crystal's (Ed.) "Cambridge Encyclopedia of Language", in particular, the section on Linguistic Typology and word order.

II. The Meyer-Hardy-Weinberg Equilibrium. This concept is the same as the textbook definition, except with one additional assumption added to the traditional five, namely, the effect that cultural evolution has upon gene frequencies, throughout subsequent generations.

III. Meyer's Optical Approximation of Anthropoid Human Eyes (MOAAHE) : A theoretical device, or rather, series of simplifying assumptions, utilized in the description of human-like vision systems, with such systems being either natural or artificial, and which follow very closely some 4500 years of empirical research and investigation into the structure and function of the human visual system. Beginning with "Le Scribe Accroupi" and other early Egyptian lenses (as described in detail by Jay Enoch, circa 1999, 2000, 2007, 2009), and such early treatises as Euclid's "Optics" (Burton, JOSA 1945; circa 300 BCE), and as further explicated by the conical sections (hyperboloid shape of a plane, constructed at a right angle to a given right cone) of Apollonius of Perga (circa 200 BCE), and the calculus of the spherical band, and cap, it can be seen within the culminating work of Rudolf Luneburg (1947), and some few other researchers at the Dartmouth Eye Institute (1919-1948), that Non-Euclidean descriptions of visual space, based upon the Euclidean defined object and image spaces, needn't be necessary for explication of such functional optical models. The mapping of points in object space, to their respective points (or, rather, "patches") in image space--and "visual space"--can take place through simple consideration of standard geometry, rectilinear propagation of pencils of light, Euclidean-type considerations of resolution, the photoreceptor counts and the historical estimates of

Osterberg and Curcio et. al. (1935 and 1991 respectively), the considerations of the pupil size, depth of field and the near triad of Alpern at al. (1961), mathematical models of human optics and accommodation, via Gullstrand (1906, 1908), Le Grand (1957) and many others in the modern tradition (Lotmar, 1971, Kooijiman, 1983, and etc., JOSA, 20th century). More precisely, Meyer's Model (MOAAHE) assumes the outline of a hyperbolic floor, where a right angle cone intersects with a planar ground, 60 degrees half field and 120 degrees full monocular field, a young near point if 12.5 cm (Eugene Hecht, "Optics" textbook, circa 2000), and a far point of accommodation, depending upon illumination levels, between 1 and 3 meters, or possibly 5 meters (Davson, 1990, and Campbell, 1954), whereupon depth of field eventually, extends to infinity, a full accommodated power for the human eye being somewhat between 60 and 73 diopters (Oyster, 1999, Davson 1990). MOAAHE assumes an eyeball height of 6 feet, and an inter-pupillary distance of 2 inches, and an eyeball radius of 12.7 mm (½ inch). It should be noted here that Smith and Atchison's (2000) standard work on the subject of mathematical models of the human eye's optics contains no reference to this model, as the MOAAHE is a heretofore unpublished model, i.e. set of assumptions and speculative theoretical idealization.

IV. Meyer's Standard Strong Artificial Intelligence Directories (MSSAID, circa October 2020 and Present) : A natural extension of information science evolution, since the I Ching binary and the Jacquard's loom, Meyer's initial insight into linguistic combinatorics (Meyer, 2003; above, **I.**) was facilitated by a Windows 10 Computer "update" (crash) in 2016. Meyer's subsequent experience with Linux boot media, the ethical principles of FSF/EFF/GNU-Linux, and with the ease of use--and code modification and code borrowing standards—along with a mild adaptation of Rodney Brooks' (1990s) subsumption architecture, inspired the next step to contemplate memory management techniques for a putative "General A.I." , or "Strong A.I.". Of further import, familiarity with the book "Programming the 6502" (Zaks, 1983) via his father's musty old New England Tech (1985; Associate's Degree) basement books, had led Meyer to join the 6502.org online forum (2013-2015) and to design a legacy 6502, four chip computer system. On these forum pages, memory management techniques and General A.I. issues were discussed, preparing him for the switch to a Linux based, low cost, Raspberry Pi Zero, "Bramble Cluster" system. Several (approximately 5 QTY) Raspberry Pi Zeros, a 4-bay RAID array "box", and 2 QTY 3 TB HDD's were acquired circa 2016-2019 to affect the project's end goals, but political considerations intervened and the work was destroyed. After a forced move on New Year's Eve, 2020, and a brief bout with cancer ("cured", circa July 2020) and the concomitant COVID social restrictions, another move, this time to Vancouver, Canada, was affected, and in this fashion, MSSAID was conceived and outlined on the tinycore, raspberry pi forum page, through the wi fi connection of a pandemic-quarantine wing of a Holiday Inn, in Cloverdale, British Columbia, in October of 2020. The basic idea was fleshed out, though subsequent research has indicated that a bramble pi cluster is not the ideal implementation, since arduinos or other faster microcontrollers have less "overhead" and "house cleaning" operations than a full Linux implementation. Tiny Core (PiCore v. 10, I think?) is still contemplated, but a switch to microcontrollers is definitely desirable, for the faster acquisition of sensory information. In short, MSSAID is a simple addition of 5 new directories to the Linux Standard Base (LSB) File Hierarchy System (FHS). Complexity beyond this simple hypothesis is contemplated, but at present, such emendations are inchoate and necessarily so; pending further research, design and development of the ideas. In summary, the standard folders, somewhat unchanged since the original Unix and Linux days, will additionally have folders labeled "vis", "aud", "tact", "olf" and "gust" for the five senses, visual, auditory, tactile, olfactory, and gustatory. A short "shebang" bash script was worked up at that time (late October, 2022) but political considerations again intervened, delayed implementation, further, and another forced move destroyed that research as Mr. Meyer fled to Hawaii, to continue his research.

V. Olfactory, Gustatory, SAWs and FOICs (2020-Present: 2030 expected completion.): Having already

worked out the efficient and effective digitization and storage of 2 (or 3) of the 5 major human sensory datasets, it should be apparent to even the most casual observer that a transducer array capable of collecting anthropoid olfactory and gustatory sensations is the obvious next step in attempting to achieve Strong A.I. With this in mind, I have given myself this decade to draw up a plan of attack, considering the approximately 400 active Human Olfactory Receptor Proteins and their many permutations and functional attributes, considering the peculiarities of Surface Acoustic Wave Device Design and the peculiarities of Fiber Optic Sensor Array bundles, and finally, considering the chemical structures of expected stimuli and their various discernment thresholds as exhibited by the human animal. Comparisons to the ancient texts of the Bhagabati Sutra and the Sushruta simhata will likely be made, but new combinatorics will be employed as well, and will be added to the larger program of memetic combinatorics, and to the linguistic typology of natural languages calculations, already in progress, as outlined above.

Work Experience:

1. Studied abroad at the Bermuda Biological Station for Research. General collection, taxonomy, and systematics of tropical marine invertebrates, with a focus on Tardigrada. 2001.

2. Temporary Employee: Pawtucket (RI) Credit Union : Mortgage pre-application filing. 2003.

3. Founded Sams Antics Inc., in the State of Rhode Island, a corporation dedicated to (a) the publishing of my unique hypotheses and theories, and (b) to the testing of those hypotheses in the furtherance of the program of Strong Artificial intelligence. Chiefly concerned with Meyer's Linguistic Law, namely, V(N^2) = S, but also concerned with two tables describing meme-gene co-evolution (and/or Lockean/Humean "ideas" and/or the "culturgens" of Lumsden and Wilson). 2004

4. Cabot Creamery, Quality Control. Studied VLSI design (Weste and Eshraghian), hardware and logic design (Wakerly), and "Scientific Charge Coupled Devices" i.e. image sensors, via Janesick, 2001) in my off hours. 2005.

5. IBM, Rapid Thermal Anneal Technician. Studied Kingslake, lens design, and general optics (Hecht) in my off hours. 2006

6. Unintentional, 3 second cameo, in the Ralph Nader movie "An Unreasonable Man"(footage from 2000--whilst an undergrad, in my junior year--Madison Sq. Garden, October 13th, 2000). Circa 2006.

7. Numerous robotic eye designs (my research circa 2003-2014, and current). Research centered on foveated imagers, following Sandini (1980) and Van der Spiegel (1989); Focal Plane Arrays (FPAs), and Focal Surface Array (FSAs) (following NASA Cassini and etc. mission imager designs, circa 1980s and 1990s); Goldberg polyhedra (Goldberg 1937 and Hart circa 2000s); various rudimentary raspberry pi, stepper motor, and nitinol "muscle wire" (Shape Memory Alloys, SMAs) experiments. Also, an in depth study of the human eye; Wertheim acuity (1897); Osterberg/Curcio's retinal cell counts (1935 and 1990, respectively); Gullstrand's optics (1909); special low index optical glasses (Hoya Glass?), crystals (Schumann 1890s, Stockbarger, 1920s-1930s), plastics (Cytop, Fluoropolymers, etc.), and liquids (Cargille standard samples); ancient optics (via Jay Enoch's 2000 analysis of "Le Scribe Accroupi"); Fiber Optic Image Conduits (FOICs) and field flatteners (via the works of Brian O'Brien, William Hicks, and Ernst von Brucke); and functional scale models of the human optical system (via Gliddon, and Schultz, 1920s and 1940s, respectively). Studied the history and philosophy of "memetics", Richard Semon, "engram", Bertrand Russel "Analysis of Mind", Golgi and Cajal's

reticular and neuron theories, and Aristotle's inchoate, though perhaps prescient, ruminations on the subject.

8. Recorded a folk-classical album (or two or three; original words and lyrics and self-recorded and self-published), guitar and vocals, circa 2012 to 2013. Limited Distribution. Approximately 300 CDs given away while collecting signatures to get on the ballot for candidate, U.S. House, State of Vermont (Independent) 2014.

9. College/Community Radio Disc Jockey, WGDR, Plainfield, 91.1 MHz. 2013-2014, and 2015-2016.

10. Ran for Congress, 2014, United States House of Representatives, Vermont, as an Independent Candidate. Debated Peter Welch, and others, on VPT. (I won the debate, but lost the election.)

11. Intensive hobbyist study of 6502 legacy systems, chip architecture, and assembly language (2000-2004; 2013-present) and newer raspberry pi, cluster computers (2015-present) and transducer integration (2003-present). Some python programming and bash scripting (2019-present), to supplement my minimal "10th grade" TI-82 programming skills (1996-1998).

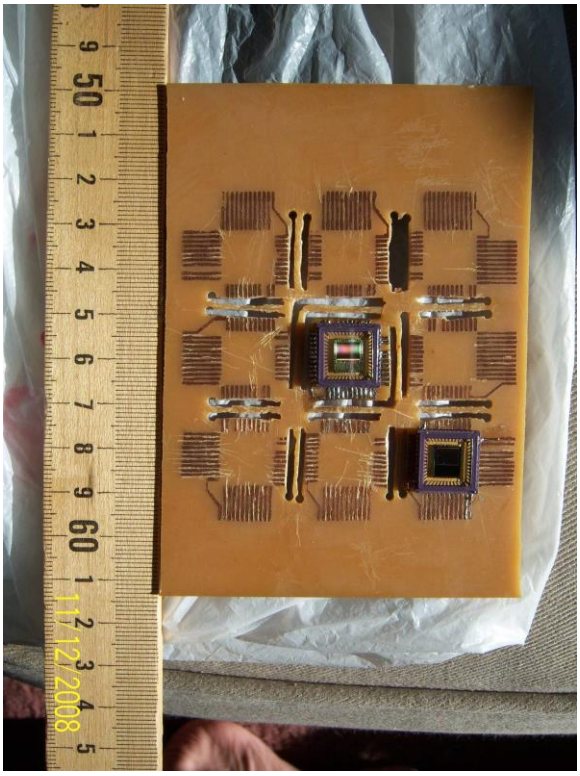
12. Moved to Canada (2020), seeking a new start and pursuing an old goal. A PhD in evolution biology, and a published book. Deported 2022, and limped to Hawaii with my some of my scientific notes and some of my scientific apparatus.

13. Second week in Canada (Sept-Oct 2020), conceived of a plan for expansion to the standard Linux operating system directory tree (LSB; FHS), and posted a plan for MSSAID (Meyer's Standard Strong Artificial Intelligence Directories) on the tiny core forum, posted under the username ladnar.

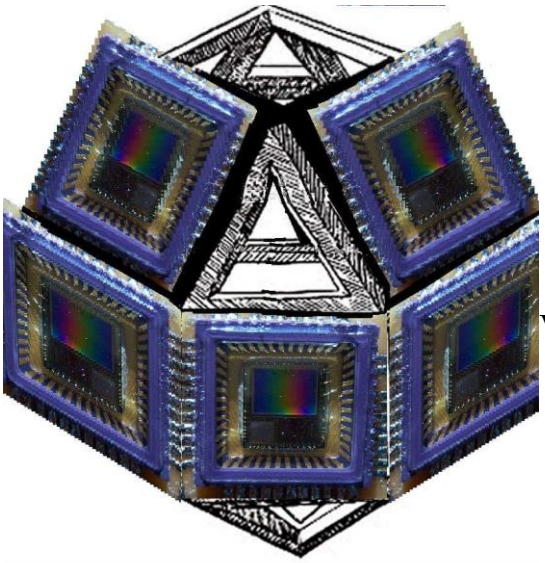
14. Design of a board game, "Landlord's Game" or, rather, "Monopoly Type" game, in the fashion of Lizzie Maggie, and/or the Parker Brothers, devoted to teaching children and teenagers the history of technology, the evolution of intellectual properties and human discoveries, and actual CPU/ALU programming techniques, in assembly or machine language. The game has three parts, first, (1) to be played as a simple "technopoly", or Monopoly type game, second, (2) with 16 Intellectual properties in a "timeline" taking the place of each of the physical properties (e.g. Park Place, Boardwalk, etc.) with modified rules to facilitate the notion of evolving timelines, and finally, (3) a game in the center of the board devoted to teaching assembly/machine language programming of CPUs of specific model, e.g. 6502. Begun circa 2015, but interrupted and taken up again in Spring and Summer of 2021 (almost finished), and largely documented on the 6502 forum. Python and Android programming of the game is being undertaken. March through August 2021 (present). Github presently still titled LadnarofLadner, but may change soon, given my move from Ladner to the Hilo, HI (October 13, 2022), greater metropolitan area. In summary, game design since 2013-2019, and second attempt 2020-present.

Ralph Nader Green Party Rally.
Madison Square Garden, 10/13/2000

Fused Silica “Cornea”, 10 mm. Diameter of Clear Aperture, 0.7 mm. Apex Thickness and R1=7mm. And R2=7 mm.
Circa 2007-2012.

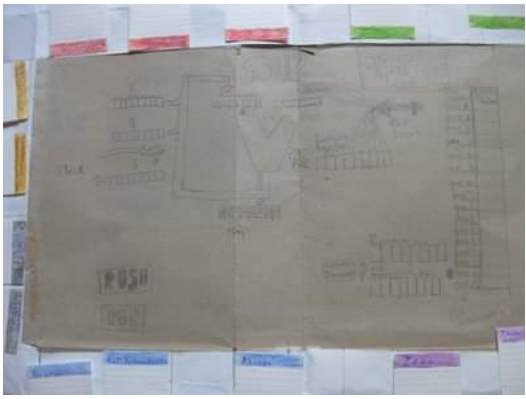


Aborted attempt at a low cost Focal Plane Array, using commercially available Kodak CMOS Sensors. Circa 2007-2012.

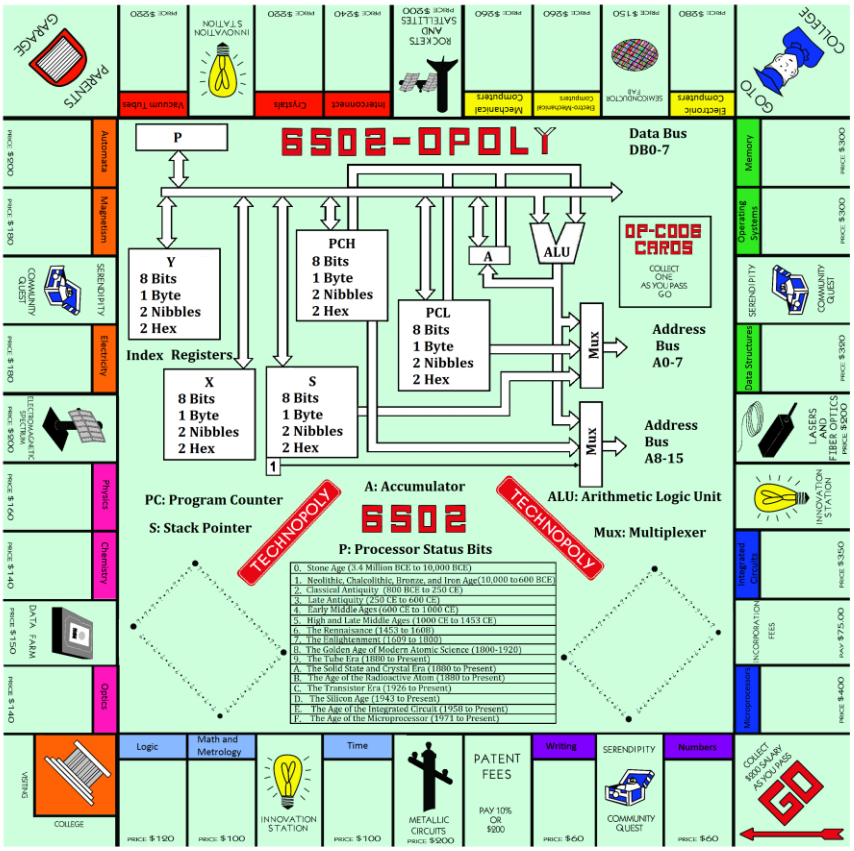


Artistic representation of inverted (Convex) Focal Surface Array. Overlaying on a Leonardo da Vinci diagram. Created circa 2013-2016.

U.S. House of Representative Debate, Vermont Public Television, 2014.



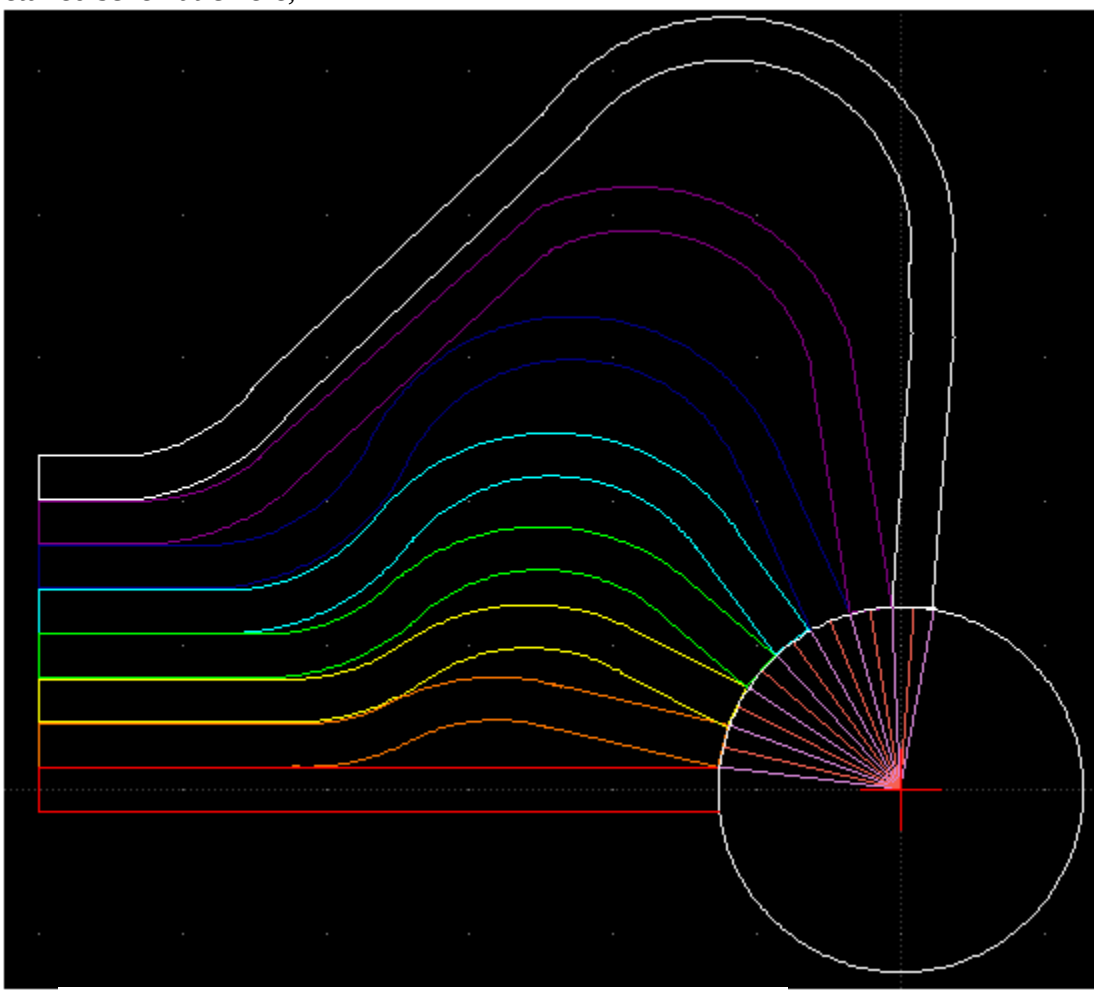
Technopoly mock-up, circa 2015-1016.

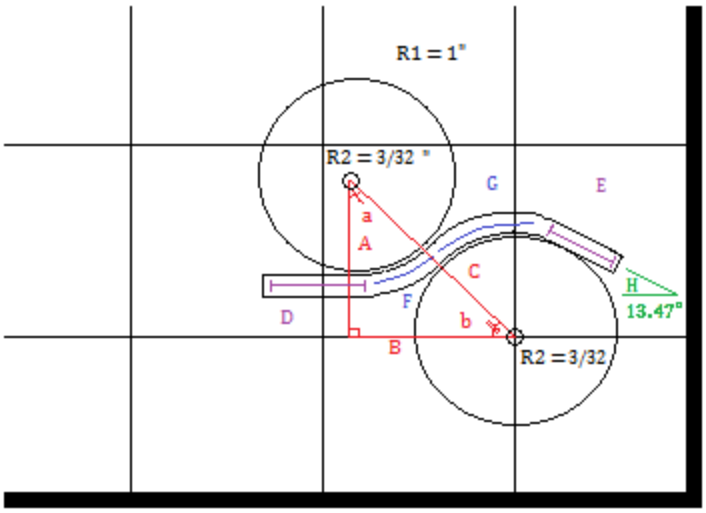


Final Technopoly Board,
as of Summer 2021, through
October.

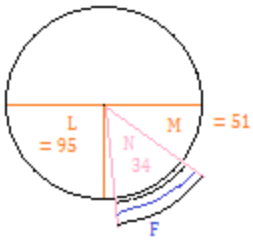
FSA with 7 cheap imagers. Glued.
Need manufacturing capabilities and open source
Datasheets. OV Omnivision Sensors;
OV-2720, or 9 similar-type, low-cost OV-WXYZ sensors; 7 QTY.

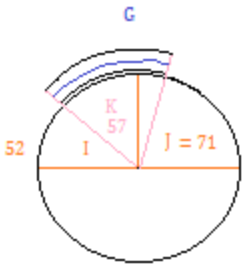
FOIC : Detailed schematic here;





Graphit Forms: Plans and such : Also; photocopy of 2011 notes and Winter 2021 Schematic. Pics below' Beside => =======>





Bibliography :

1. Meyer, Randall Unpublished, “Meyer’s Linguistic Law”, incomplete PhD Thesis. Began 2003, continuing, as the book is written, circa 2024-present.
2. Chomskyian Linguistics (1957), syntactic Structures
3. David Crystal’s (Ed.) “Cambridge Encyclopedia of Language” pgs. “Linguistic Typology”
4. Jay Enoch, circa 1999, 2000, 2007, 2009),
5. Euclid's "Optics" (Burton, JOSA 1945; circa 300 BCE), a
6. Apollonius of Perga (circa 200 BCE),
7. Rudolf Luneburg (1947),
8. Dartmouth Eye Institute (1919-1948), t
9. Osterberg , 1935
10. Curcio et. al. 1991
11. Alpern at al. (1961),
12. Gullstrand (1906, 1908),
13. Le Grand (1957)
14. (Lotmar, 1971,
15. Kooijiman, 1983,
16. JOSA, etc. 20th century).
17. Eugene Hecht, “Optics” textbook, circa 2000),
18. Davson, 1990,
19. Campbell, 1954), w
20. Oyster, 1999,
21. Davson 1990).
22. Smith and Atchison’s (2000)
23. Randall Meyer, uinpublished, MSSAID stated, tiny Linux forum (2020)
24. Rodney Brooks’ (1990s) subsumption architecture,
25. “Programming the 6502” (Zaks, 1983)
26. Randall Meyer, unpublished, 6502.org online forum (2013-2015)
27. Randall Meyer, unpublished, github Olfactory Project (2023-present)